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Lesson 4: Classification and Evaluation Metrics

Technologies for Artificial Intelligence

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Before we start

- Exercise 3, which we discuss now
- Ridge, Lasso, and the choice of the penalty strength

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Today: from a number to a decision

- Logistic regression: the model and why the cost had to change
- The confusion matrix, and why accuracy lies
- Precision, recall, F1, and the threshold nobody chose
- Receiver operating characteristic (ROC) curves, and where they mislead
- Choosing a threshold from money

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The question has changed

- Lesson 3: given a house, **how many euros?**
- Lesson 4: given a disk drive, **will it fail?**
- Answer with a probability: $0 \leq P(y = 1 \mid x) \leq 1$
- $y = 1$ if the drive failed, $y = 0$ if not; x holds its six counters

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The fleet

- **8,000 disk drives** in a data centre
- Six SMART counters each: Self-Monitoring, Analysis and Reporting Technology
- Reallocated sectors, spin retries, read and seek error rates, temperature, hours in service
- Label: did it fail within thirty days?
- `train_test_split(..., stratify=y)`: 6,000 to train, 2,000 to test

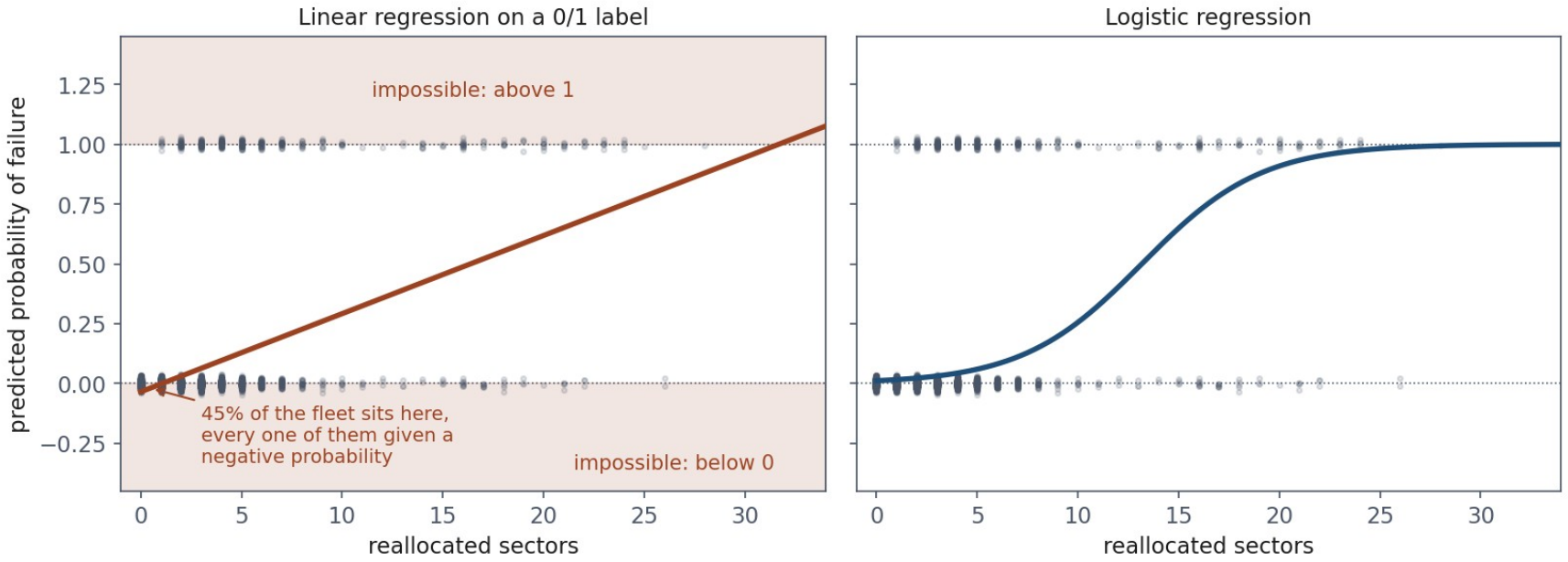
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306

- **306 of the 8,000 drives failed.** 3.8% of the fleet
- Which means 96.2% did not

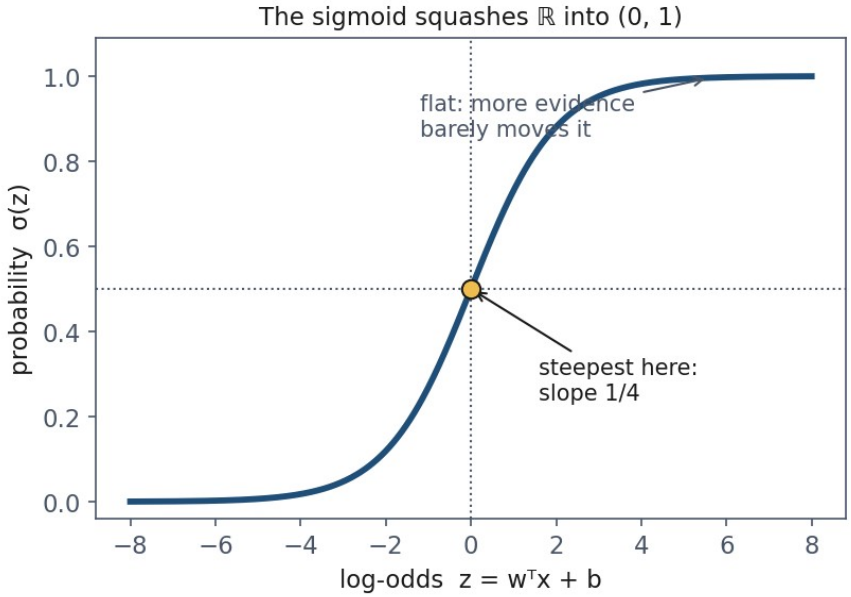
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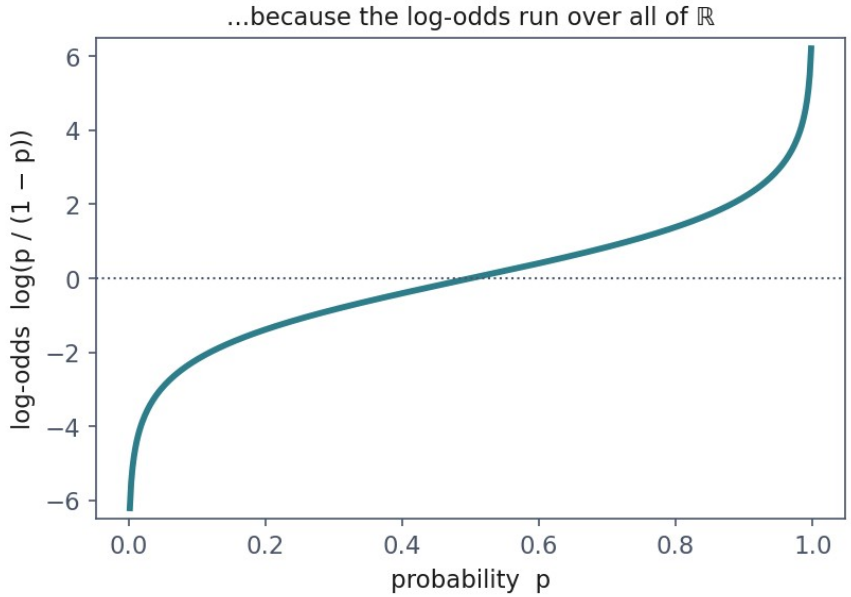
What goes wrong if we just fit a line



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The sigmoid





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Why the log-odds

- A line gives any number; a probability, only [0, 1]
- Odds, $p \div (1 - p)$: no ceiling, from 0 to ∞
- Log of the odds: no floor either, $-\infty$ to $+\infty$
- $p = 0.01, 0.5, 0.99 \rightarrow$ log-odds $-4.60, 0, +4.60$
- Bonus: coefficients multiply the odds (slide 13)

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Evidence multiplies the odds

One reading, 6 times more common among failing drives	Drives at 50% risk	Drives at 95% risk
1,000 such drives: will fail / stay healthy	500 / 500	950 / 50
Those showing the reading: 60% of failing, 10% of healthy	300 / 50	570 / 5
Risk after the reading: failing ÷ all showing it	300 ÷ 350 = 86%	570 ÷ 575 = 99.1%
Adding 0.10 to the risk instead	60%	105%: impossible
Odds = failing ÷ healthy, before and after	1, then 300 ÷ 50 = 6: × 6	19, then 570 ÷ 5 = 114: × 6

Log-odds of p = P(y = 1 | x): a line

$$\log \frac{p}{1-p} = w^\top x + b \quad \Longrightarrow \quad p = \frac{1}{1 + e^{-(w^\top x + b)}}$$

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The model, in one line

$$P(y = 1 \mid x) = \sigma(w^\top x + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}$$

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What the coefficients mean

- The model is linear in the log-odds, log(p ÷ (1 − p)), so `exp(coefficient)` multiplies the **odds**, p ÷ (1 − p)
- Features standardised, so one unit = one standard deviation
- A coefficient of 1.80 means **six times the odds** of failing
- In code: `make_pipeline(StandardScaler(), LogisticRegression())`

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The truth, and what we recovered

Feature	True	Fitted	Odds ×
reallocated_sectors	1.80	1.66	5.28
spin_retry_count	1.15	1.17	3.22
read_error_rate	0.85	0.87	2.39
temperature_c	0.45	0.47	1.59
power_on_hours	0.35	0.38	1.46
seek_error_rate	**0.00**	**0.02**	**1.03**

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The intercept is the average drive’s chance

- Fitted intercept: **−6.09**
- σ(-6.09) = 0.0023: a 0.2% chance for a drive average on every counter
- The fleet fails at 3.8%: the failures come from the few drives far above average

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Fitting means minimising something, but what?

- Squared error worked for lesson 3
- Why not just apply it to the sigmoid’s output?

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Step 1: each drive gets a p

Drive	What happened	y	Setting 1: $p = \sigma(w^T x + b)$, the P(fail)
A	failed	1	0.8
B	stayed healthy	0	0.1
C	stayed healthy	0	0.2

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Step 2: how likely was the outcome?

Drive	y	p	Probability of what happened: p if y = 1, 1 - p if y = 0
A	1	0.8	0.8
B	0	0.1	$1 - 0.1 = 0.9$
C	0	0.2	$1 - 0.2 = 0.8$
All three: multiply			$0.8 \times 0.9 \times 0.8 =$ **0.576**

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Step 2 as formulas: the likelihood

$$P(y \mid x) = p^{\,y}\,(1-p)^{1-y}$$

$$\mathcal{L}(w,b) = \prod_{i=1}^{m} \left(p^{(i)}\right)^{y^{(i)}} \left(1-p^{(i)}\right)^{1-y^{(i)}}$$

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Step 3: two settings, the likelier wins

Drive	y	Setting 1: p	Setting 1: probability of what happened	Setting 2: p	Setting 2: probability of what happened
A	1	0.8	0.8	0.3	0.3
B	0	0.1	0.9	0.6	0.4
C	0	0.2	0.8	0.5	0.5
Likelihood: the product			**0.576**		**0.060**

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Step 4: minus the log, a cost

Drive	y	Setting 1: −log of probability of what happened	Setting 2: −log of probability of what happened
A	1	−log 0.8 = 0.22	−log 0.3 = 1.20
B	0	−log 0.9 = 0.11	−log 0.4 = 0.92
C	0	−log 0.8 = 0.22	−log 0.5 = 0.69
Sum = −log of the likelihood		**0.55**	**2.81**

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Cross-entropy, also called log loss

$$L = -[\, y \log p + (1-y) \log(1-p)]$$

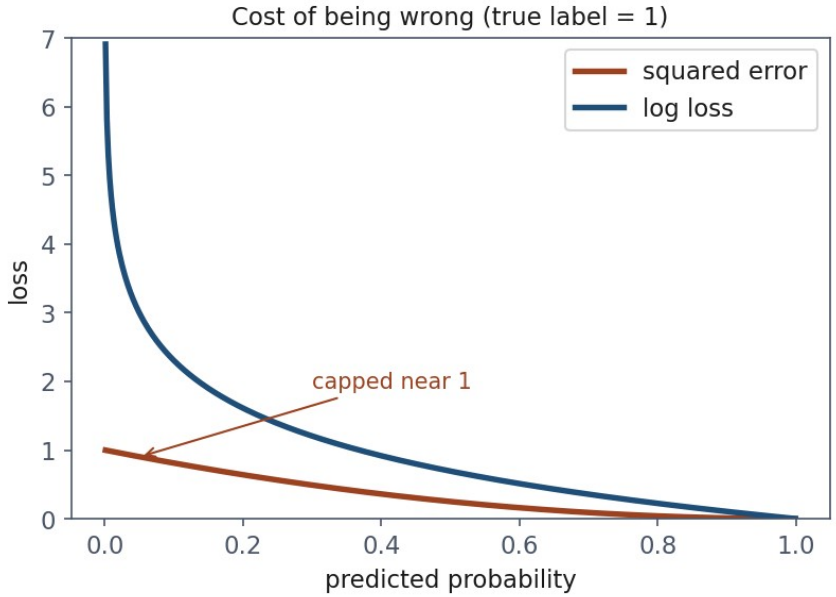
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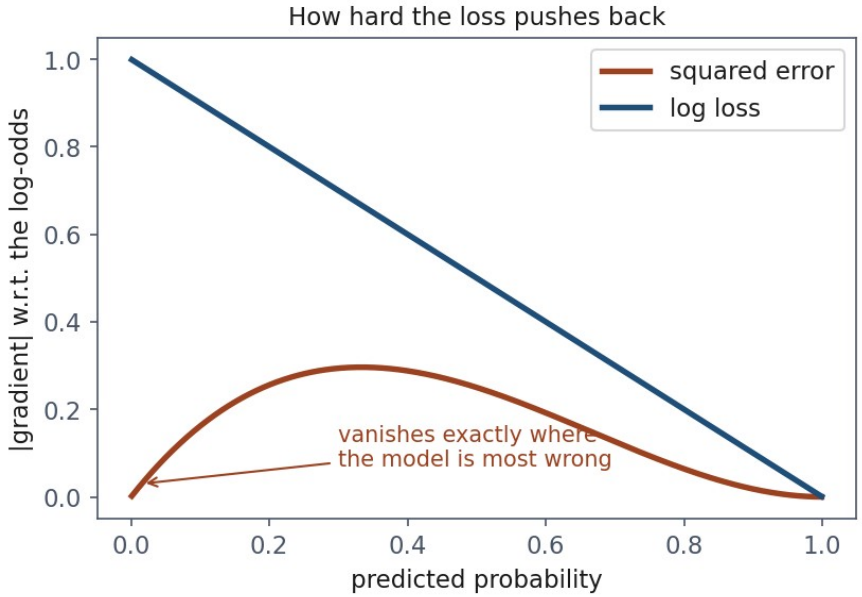
Lesson 3’s gradient, $\hat{y} = \sigma(w^T x + b)$

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^{m} (\hat{y}^{(i)} - y^{(i)}) x_j^{(i)}$$

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So why not squared error?





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And it is convex, which squared error is not

- Log loss is **convex** in the weights, so any local minimum is global
- Squared error through a sigmoid is not

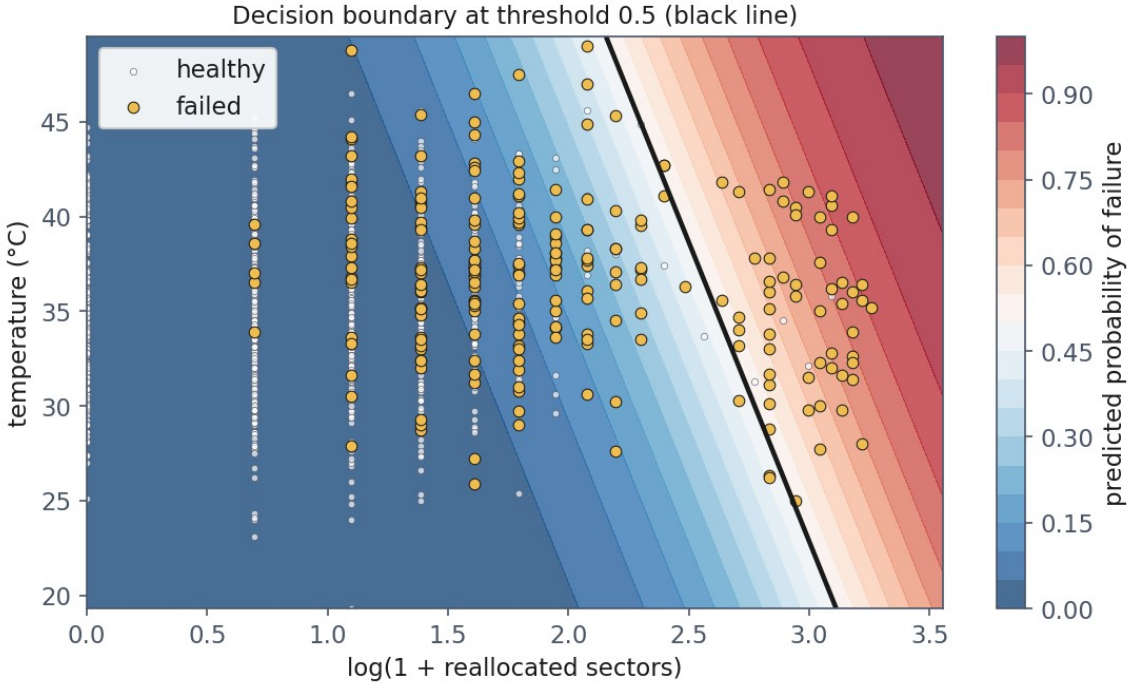
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Notebook 1, live

section, and its slides	you call (→ the number to come back with)
a line on a 0/1 label, 7	`LinearRegression` → 3,606 negative probabilities
odds, and why not squared error, 8-25	NumPy only: the sigmoid, the two losses, their gradients
gradient descent, 23	`train_test_split`, `StandardScaler`, fifteen lines of NumPy → log loss 0.6931 to 0.0692
the decision boundary, 27	`make_pipeline(StandardScaler(), LogisticRegression())` on two features
the truth, recovered, 14-15	`LogisticRegression`, `.coef_`, `.intercept_` → seek_error_rate 0.02, intercept −6.09

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The decision boundary



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Break

- Ten minutes

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Two models

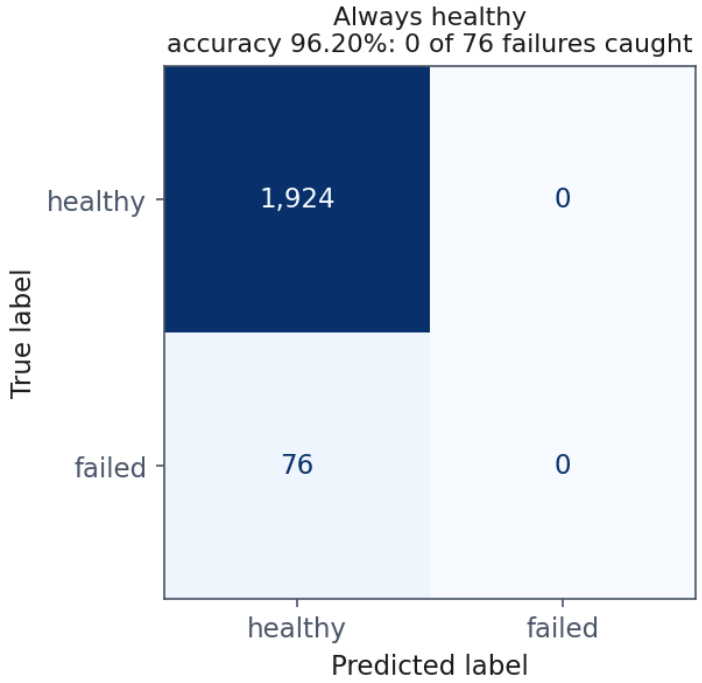
Model	Accuracy	Failures caught
Predict “healthy” for everything	**96.20%**	**0 of 76**
Logistic regression, threshold 0.5	97.70%	43 of 76

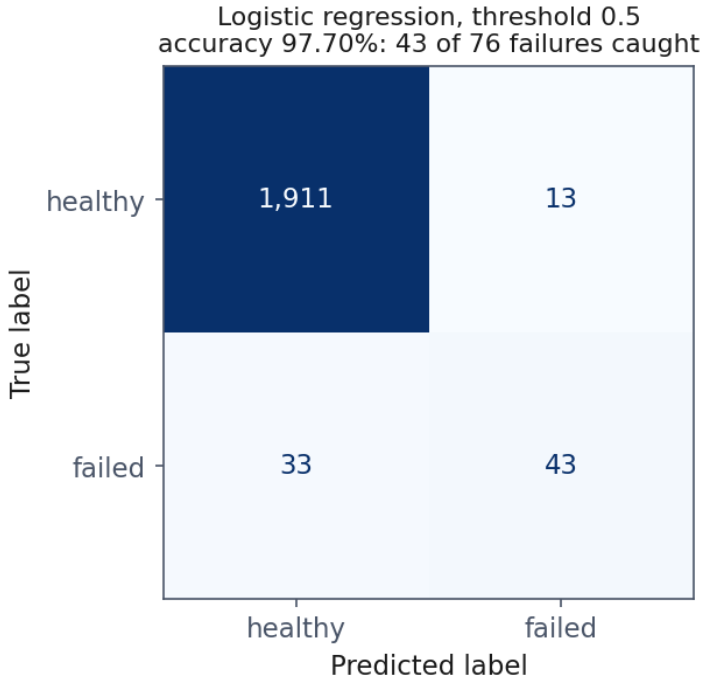
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Why everyone makes this mistake once

- Accuracy is the fraction of predictions that were right
- On a **balanced** problem that is exactly what you want
- The interesting class is almost always the **rare** one

Four numbers instead of one





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The four outcomes, and what they cost

- **False positive**: we replace a healthy drive. A technician’s hour, €140
- **False negative**: a drive dies in service. Callout, rebuild, risk, €2,600

Precision and recall: look at the denominator

Precision = TP / (TP + FP)
"of the alarms we raised"

	predicted healthy	predicted failing
actually healthy	TN 1911	**FP** 13
actually failed	FN 33	**TP** 43

Recall = TP / (TP + FN)
"of the drives that failed"

	predicted healthy	predicted failing
actually healthy	TN 1911	FP 13
actually failed	**FN** 33	**TP** 43

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Our model, in those terms

- True and false positives and negatives: TP, FP, FN, TN
- Precision = TP ÷ (TP + FP) = 43 ÷ 56 = **0.768**: of the drives flagged
- Recall = TP ÷ (TP + FN) = 43 ÷ 76 = **0.566**: of the drives that failed
- Specificity = TN ÷ (TN + FP) = 1,911 ÷ 1,924 = 0.993
- In code: `precision_score`, `recall_score`

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F1: why the harmonic mean

Model	Precision	Recall	Arithmetic mean	Harmonic, F1 = 2PR ÷ (P + R)
Flag every drive	0.038	1.000	0.519	**0.073**

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Read the minority class, not the average

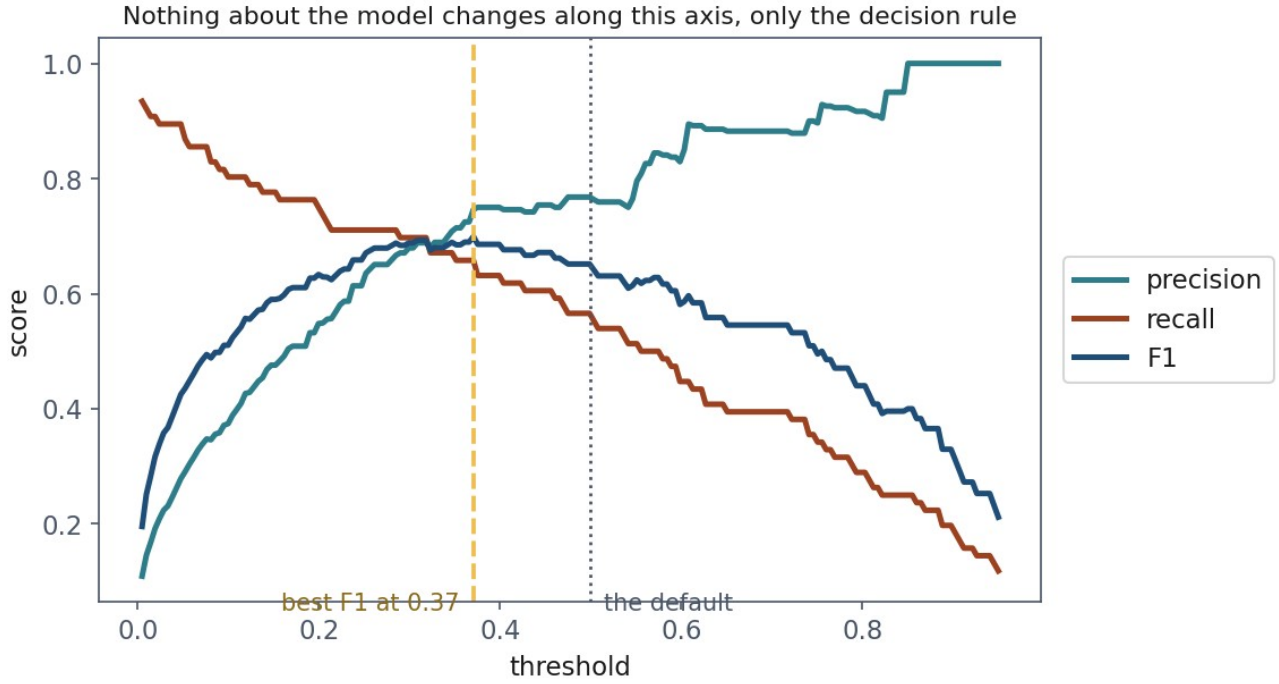
- `weighted avg` F1: **0.975** (dominated by the healthy class)
- `macro avg` F1: **0.820** (both classes get an equal say)
- In code: `classification_report(y_test, y_pred)` prints both rows

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Notebook 2, live

section, and its slides	you call (→ the number to come back with)
the number that should worry you, 29	`accuracy_score` → 96.20% for “always healthy”
the confusion matrix, 31	counts by hand, then `confusion_matrix`, `ConfusionMatrixDisplay` → TP 43
precision and recall, 33-36	`precision_score`, `recall_score`, `f1_score`, `classification_report`
the threshold sweep, 38-39	`predict_proba`, then the metrics at each threshold
the precision-recall curve, 40	`precision_recall_curve`, `average_precision_score` → 0.717 against 0.038

The threshold is a choice nobody made



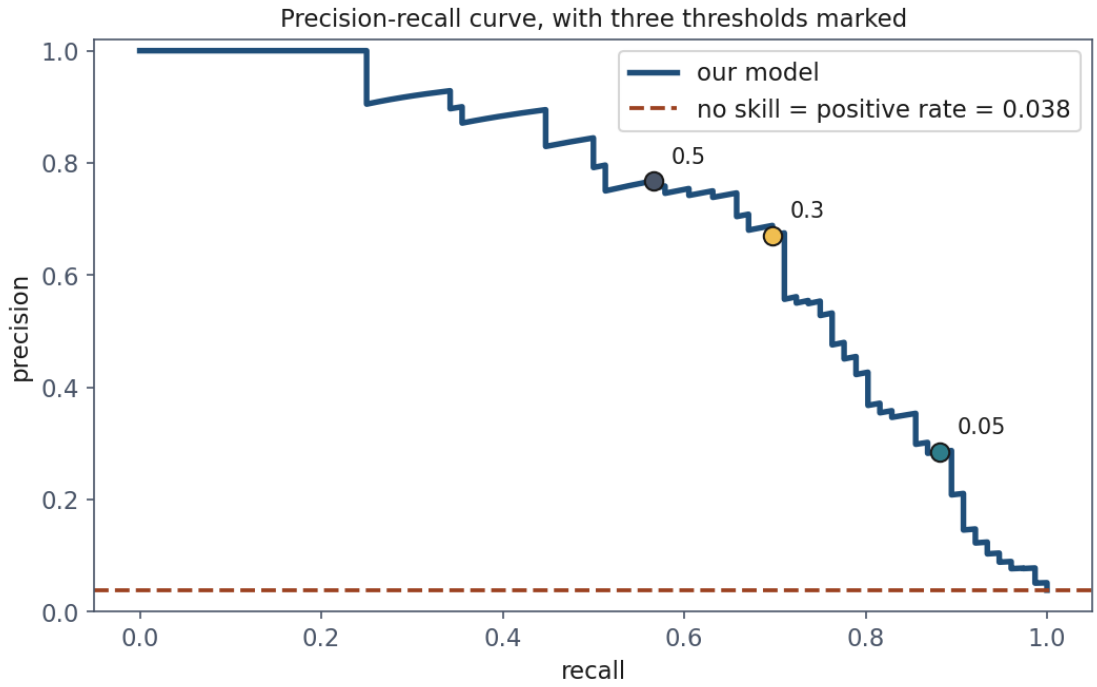
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What the sweep actually costs

Threshold	TP	FP	FN	Precision	Recall	Accuracy
0.02	69	284	7	0.195	0.908	0.855
0.10	61	102	15	0.374	0.803	0.942
0.30	53	25	23	0.679	0.697	0.976
0.50	43	13	33	0.768	0.566	**0.977**
0.90	14	0	62	1.000	0.184	0.969

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The precision-recall curve



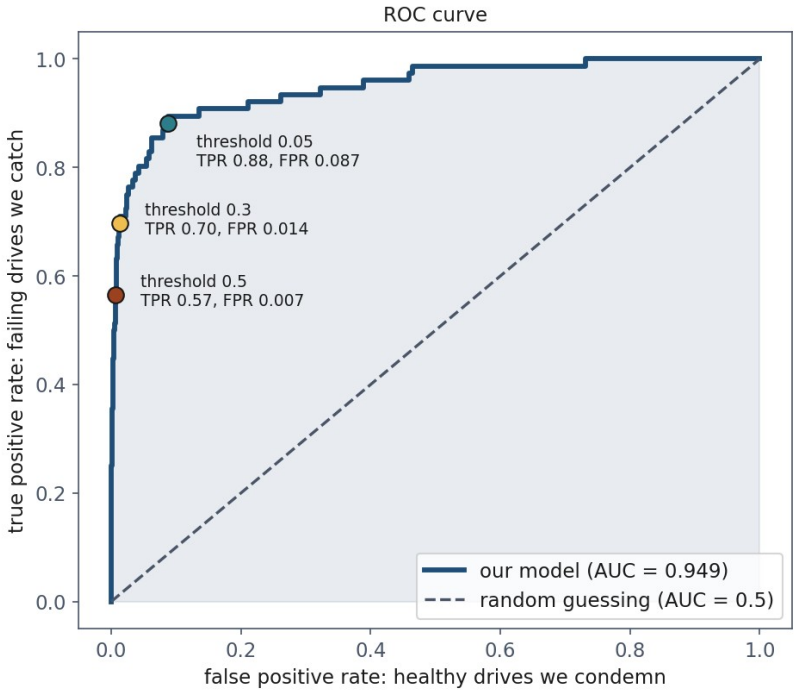
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Comparing models without choosing a threshold

- We keep evaluating at one threshold, but which?
- What if we score the whole **ranking** instead?

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The ROC curve



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What the area means

- **Area under the curve (AUC) = the probability that a random failing drive scores above a random healthy one**
- Ours: 0.949
- In code: `roc_auc_score(y_test, scores)`

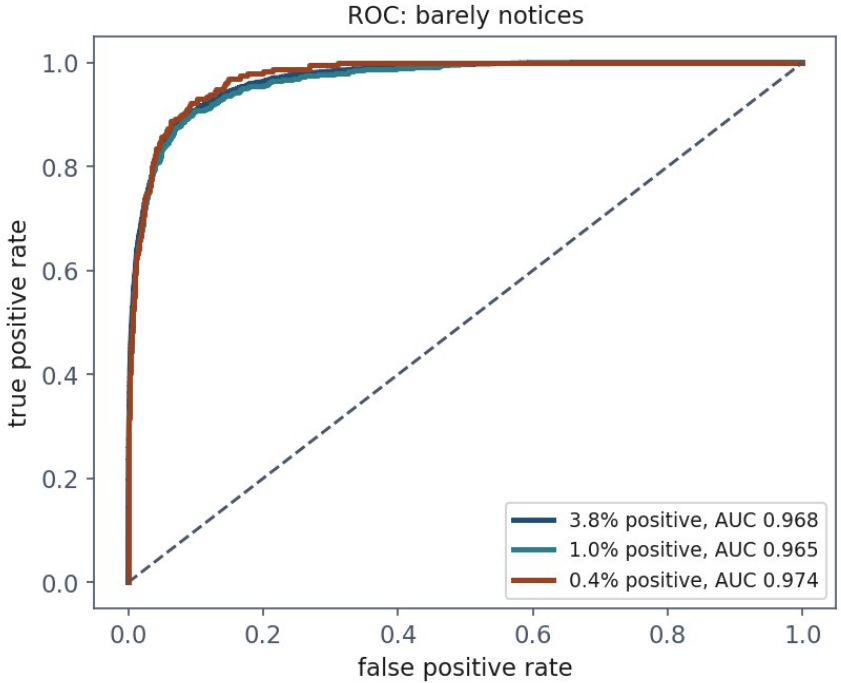
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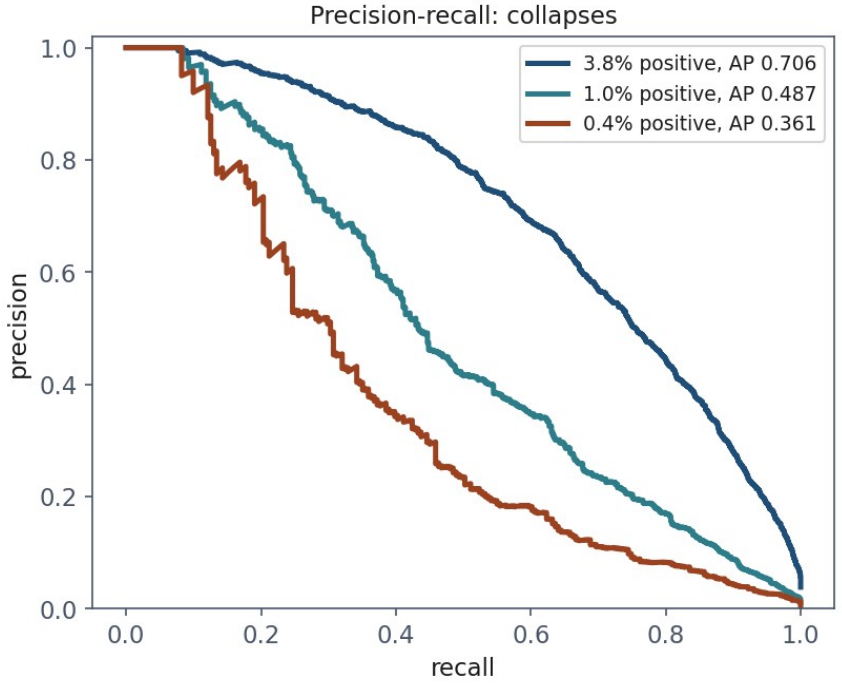
What AUC does not tell you

- Divide every predicted probability by 10
- AUC is **unchanged**; every decision changes

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Where ROC misleads





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Why the two disagree

- False positive rate (FPR) divides by **all healthy drives**: an enormous, growing denominator
- Precision divides by **the alarms raised**, where the false ones dominate

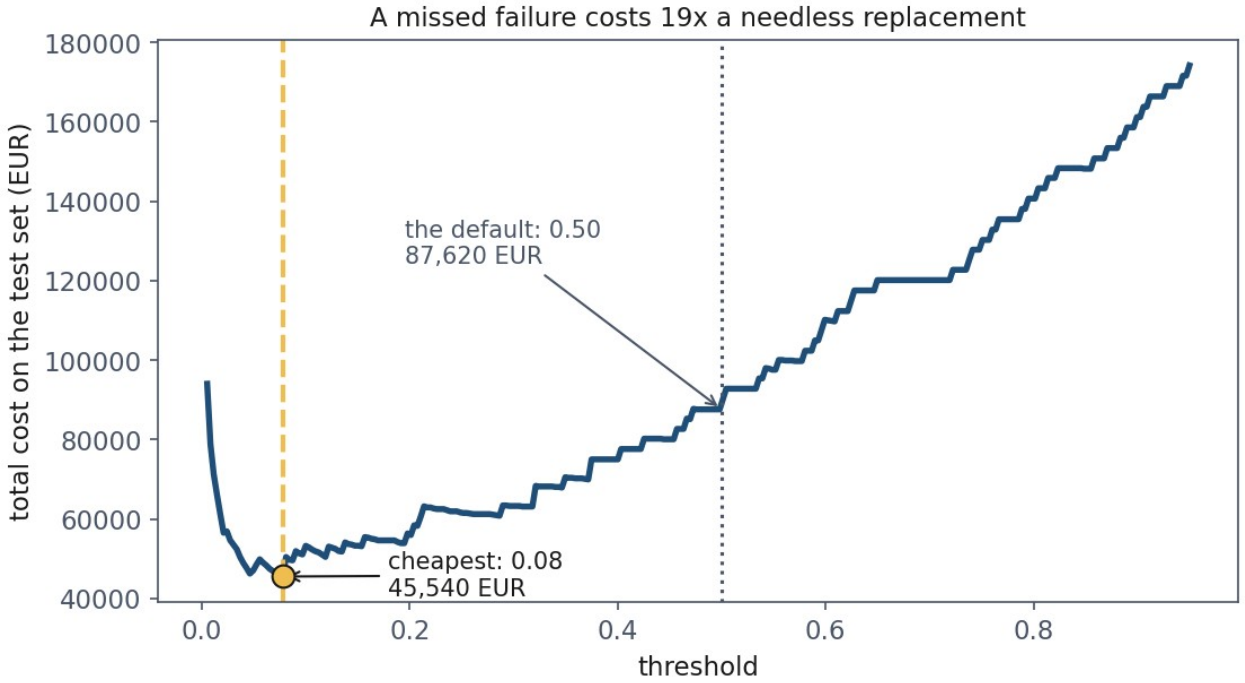
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The rule

- On an imbalanced problem, report the **precision-recall curve alongside AUC**
- Never instead of it, and never AUC alone
- Its area is the **average precision**, `average_precision_score`: 0.717 here, against a floor of 0.038

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Choosing the threshold from money



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And it has a closed form

$$t^* = \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{140}{140 + 2600} = 0.051$$

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The caveat you must state

- We chose the threshold **by looking at the test set**
- That is precisely what lesson 1 forbade

Class weights: the same lever, pulled earlier

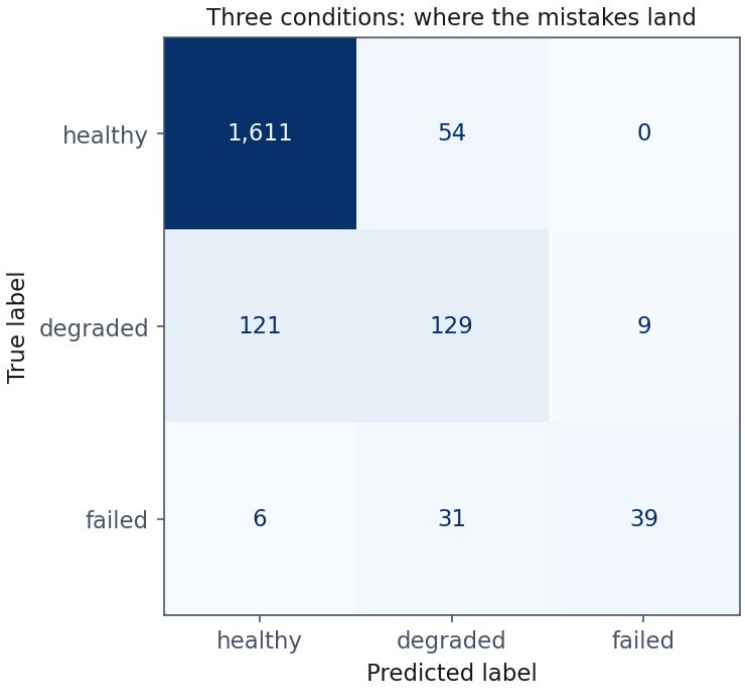
Model	Recall	Precision	AUC	Cost
Plain, threshold 0.50	0.566	0.768	0.949	87,620
Plain, threshold 0.08	0.855	0.349	0.949	45,540
Balanced class weights, 0.50	0.895	0.249	**0.950**	49,500

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Notebook 3, live

section, and its slides	you call (→ the number to come back with)
the ROC curve by hand, 42	`roc_curve`, `roc_auc_score` → 0.9493
what AUC means, 43	200,000 random (failing, healthy) pairs → 0.9488
the rare-positive trap, 45-46	`average_precision_score` beside `roc_auc_score` → AP 0.706 to 0.361, AUC unmoved
the threshold from costs, 48-50	`confusion_matrix` at each threshold → 45,540 EUR at 0.08, against 87,620 at 0.50
class weights, 51	`LogisticRegression` with balanced class weights → AUC 0.950
three classes, 53-54	`classification_report` → macro F1 0.707, weighted 0.883

More than two classes



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Macro against weighted, one last time

- macro avg F1: **0.707**
- weighted avg F1: **0.883**

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What to take away

- **96.20% accuracy, nothing caught.** Accuracy measures the majority class
- Precision is about the **alarms**, recall is about the **failures**
- The **threshold is a decision**, and it has a formula
- **AUC measures ranking** and hides imbalance

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Homework

- **Exercise 4**, discussed at the start of **Friday 6 November**
- Build a classifier, choose a threshold from a stated cost ratio, defend it

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